Paleointensity study of Early Miocene lavas from Pingzhuang, Inner Mongolia, China

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[1] A paleomagnetic and geochronologic investigation has been conducted on 17 lava flows from Inner Mongolia, north China. Radiometric K-Ar dating experiments give an age between 23 and 22 Ma for these volcanic units. Rock magnetic measurements of thermomagnetic curves and hysteresis loops show the primary carrier of remanence to be pseudo-single domain titanomagnetite. Stepwise thermal demagnetization isolated a well-defined direction of characteristic remanent magnetization (ChRM) in all studied lava flows with a mean direction of D/I = 359.8°/62.2° (α_{95} = 3.8°). Fifty samples from 11 lavas were preselected on the basis of thermomagnetic analyses to carry out Thellier-Thellier paleointensity experiments with systematic partial thermoremanent magnetization (pTRM) checks. Twenty-three samples from 7 lavas yield reliable paleointensity estimates with a mean virtual dipole moment (VDM) of $(3.9 \pm 0.9) \times 10^{22}$ Am^2, about half the strength of the modern-day field. *INDEX TERMS:* 1519 Geomagnetism and Paleomagnetism: Magnetic mineralogy and petrology; 1521 Geomagnetism and Paleomagnetism: Paleointensity; 1522 Geomagnetism and Paleomagnetism: Paleomagnetic secular variation; 1540 Geomagnetism and Paleomagnetism: Rock and mineral magnetism; 1560 Geomagnetism and Paleomagnetism: Time variations—secular and long term. **Citation:** Shi, R., R. Zhu, Y. Pan, G. Shi, and K. A. Hoffman, Paleointensity study of Early Miocene lavas from Pingzhuang, Inner Mongolia, China, *Geophys. Res. Lett.*, *29*(21), 2026, doi:10.1029/2002GL015990, 2002.

1. Introduction

[2] The past several years has seen a growing interest in documenting the variations in the strength of the geomagnetic field over geologic time, inputting that is crucial to a complete understand of the workings of Earth's dynamo [*Prévot et al.*, 1990; *Zhu et al.*, 2001]. However, large temporal gaps exist in the paleointensity database. Given this overall global scarcity of data for much of Early Miocene and, in particular, the lack of data from eastern Asia, we present a study of absolute paleointensity on the Hannuoba basalt flows at Pingzhuang in Inner Mongolia, China, determined by way of the *Thellier-Thellier* approach [1959] modified by *Coe* [1967].

2. Geological Setting and Sampling

[3] Pingzhuang (42.0°N, 119.2°E) lies in eastern Inner Mongolia in the northern part of China. The lava flows at Pingzhuang, known as the Hannuoba Formation, have been found to be Miocene in age largely on the basis of plant fossils identified by local geologists [*Zhao et al.*, 1994]. The lavas have been largely undisturbed since their eruption as they are almost horizontal and appear to contain no major faults. A total of seventeen lava flows (named as, for example, LU01, LU17) was sampled using a gasoline-powered diamond drill. At least seven cores (named as, for example, LU01-1, LU01-7) were drilled from each flow, each core orientated *in situ* using a sun compass and/or a magnetic compass. In the laboratory the cores were cut into at least two standard-sized specimens for palaeomagnetic examinations.

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3. Radiometric Dating

[4] Potassium-argon (K-Ar) age determinations were performed on fresh volcanic rock samples crushed and sieved to a size range between 0.20–0.28 mm. Argon gas was analyzed using the isotope dilution technique with a 99.98% pure ^{38}Ar spike on a RGA10 mass spectrometer connected to a purification and extraction system in the isotope geochemical laboratory of Department of Geology, Peking University. The spike was calibrated and corrected with standard argon minerals: ZBH-25, whose real age is 131.7 Ma and this K-Ar age determination gives value of 129.6 Ma. The decay and abundance constants used in the determinations are $\lambda\beta- = 4.962 \times 10^{-10}$ a^{-1}; $\lambda\varepsilon = 0.581 \times 10^{-10}$ a^{-1}; ^{40}K/K = 1.167×10^{-4} mol^{-1}. The analytical data and the calculated ages, ranging from 23–22 Ma, are reported in Table 1.

4. Laboratory Experiments and Results

4.1. Rock-Magnetic Investigations

[5] Rock-magnetic investigations were conducted at the Palaeomagnetism Laboratory of the Institute of Geology and Geophysics (IGG-CAS), in order to assess the reliability of the palaeomagnetic recordings. Using a Micromag 2900 alternating gradient magnetometer, magnetic hysteresis measurements were performed on a small chips (~30 mg) taken from a sample from each of the 17 lavas up to a maximum field of 1.5 T. When plotted on a Day diagram [*Day et al.*, 1977], the ratios (Figure 1a) fall within the pseudo-single-domain (PSD) size range, indicating either PSD grains or a mixture of single domain and multidomain grains [*Goguitchaichvili et al.*, 2001a].

Table 1. Age Determinations Obtained From the LU Section

Sample name	K(%)	Mass of sample(g)	^{40}Ar 10^{-11}mol/g	^{40}Ar %	^{38}Ar 10^{-11}moles	^{40}Ar/^{38}Ar	^{38}Ar/^{36}Ar	Age (Ma)
LU03-8	0.696	0.2161	2.730	60.53	2.003	0.4869 ± 0.0068	1428.9 ± 2.0	22.5 ± 0.6
LU05-7	0.70	0.2300	2.806	69.22	2.065	0.4517 ± 0.0034	1925.1 ± 0.6	23.0 ± 0.5
LU06-6	0.692	0.2116	2.731	60.86	2.006	0.4737 ± 0.0096	1477.6 ± 2.2	22.6 ± 0.8
LU15-2	0.712	0.1858	2.758	68.42	2.005	0.3738 ± 0.0025	2231.1 ± 0.4	22.2 ± 0.3

[6] Strong field thermomagnetic analyses were carried out on samples from each lava flow with a MMVFTB. Employing a heating and cooling rate of 3°C min^{-1} the results of these analyses can be generally divided into two behaviors, types A and B (Figure 1b). Type A samples have a distinct Curie temperature, which is determined as the intersection of the descending curve with the tangent to the subsequent paramagnetic tail, near 560–585°C, indicative of nearly pure magnetite, and show at most a minor amount of irreversibility of the heating and cooling curves. Such thermomagnetic behavior suggests that negligible alteration of the magnetic mineralogy had occurred during heating. Specimens from eleven of the seventeen lava flows fell into the Type A behavior category and were selected for paleointensity investigation. Type B samples, however, display strongly irreversible behavior were removed from further consideration for paleointensity determinations.

4.2. Paleodirectional Analyses

[7] Measurements of remanent magnetization were made on a JR5a spinner magnetometer situated in a field-free space (<300 nT). Natural remanent magnetization (NRM) intensity varied from 0.06 to 2.6 A/m with a mean value of 1.02 A/m. Stepwise thermal demagnetization consistently isolated a well-defined direction of ChRM in each lava. Specifically, thermal demagnetization was carried out using a Magnetic Measurements furnace with a residual magnetic field less than 10nT. Thermal steps ranged from 25°C to 50°C to a peak temperature of 585°C. Generally, a small secondary component, probably of viscous origin, was present and easily removed by 150°C (Figure 2). A stable ChRM was obtained from majority of samples (105 out of 116) after demagnetization to a peak temperature of only 200°C. Lava-mean directions and statistical parameters were calculated for each flow assuming a Fisherian distribution. The results show that all 17 flows possess normal polarity with an average direction, D/I = 359.8°/62.2° (k = 89.9, α95 = 3.8°). The calculated palaeomagnetic pole position, Plat = 88°, Plong = 120°, with A95 = 5.1°, is consistent with the published data from the North China Block [*Zhao et al.*, 1994], indicating that no displacement or rotation occurred at the sampling site since Early Miocene. On the other hand, the studied lava sequences might be erupted in a relative short time interval, which could be deduced from radiometric dating and low angular standard deviation of paleodirectional results.

4.3. Determination of Paleointensity

[8] Based on the thermomagnetic analyses, 50 type A samples obtained from 11 lavas were selected for paleointensity determination. *Coe*'s [1967] modification of the *Thellier and Thellier* [1959] double heating paleointensity method was employed for this study. A laboratory field of 50μT was applied during both the heating and cooling cycles. Argon gas was circulated slowly through the furnace during each heating-cooling cycle in an attempt to suppress possible oxidation of the titanomagnetite grains. Also, a few milligrams of activated charcoal, a buffer to any oxidation/reduction of magnetite, was introduced into the furnace prior to each run [*Laj and Kissel*, 1999]. Following heatings where the peak temperature was >200°C, pTRM checks were systematically performed in order to detect any possible alteration of the magnetic mineralogy. The highest temperatures used in the Thellier experiment is 520°C.

[9] Examples of the paleointensity results in NRM-TRM diagrams are shown in Figure 2. The NRM remaining after each thermal run is plotted against the corresponding TRM acquired in the applied laboratory field at the same peak temperature. In an ideal diagram, the NRM/TRM points fall onto a straight line, the slope of which gives the ratio of the laboratory field to the paleofield. Data was utilized in the determination of the paleointensity so long as the NRM-TRM plot remained linear and that the pTRM-check remained positive. A given pTRM-check was considered negative if its value differed from the initial value associated with that peak temperature by more than 3 percent. It should be noted that the NRM-TRM linearity breaks at 460°C for some samples, while pTRM checks are still positive after heating to 500°C or even higher. We propose that some alterations at high temperatures caused by MD grain-growth are probably responsible for this phenomenon. Of the 50 samples involved in this study, 23 samples from 7 lavas yielded reliable paleointensity results. The quality of each paleointensity determination was evaluated using the statistical parameters developed by *Coe et al.* [1978] (Table 2). As can be seen in Table 2, the paleointensity results mostly

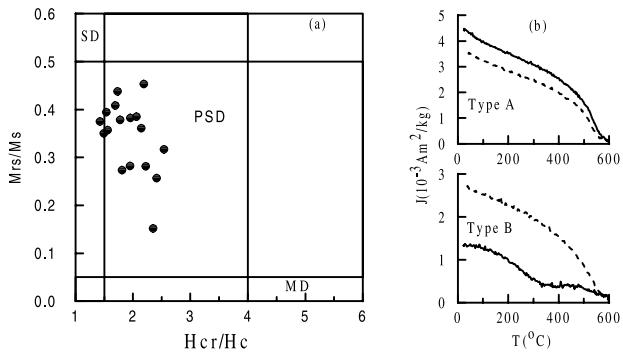


Figure 1. (a) Hysteresis ratios for Pingzhuang samples, (b) Two characteristic curve types (A and B) of thermomagnetic curves from different lava flows of Pingzhuang. Solid and dashed line stands for the heating and cooling curves, respectively.

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Table 2. Paleointensity Results

Lava	N	Dec	Inc	α_{95}	Sample	n	T_{min}–T_{max}	f	g	q	$F \pm \sigma$ (μT)	Fa $\pm \sigma_F$ (μT)	VDM $\pm \sigma_v$ ($\times 10^{22}$ A m^2)
Lu03	6	352.7	56.3	2.3	lu03-4	10	100–500	0.73	0.88	32.7	24.3 ± 0.5	22.2 ± 1.5	4.0 ± 0.3
					lu03-5	6	300–500	0.63	0.80	28.2	18.3 ± 0.3		
					lu03-6	6	300–500	0.54	0.79	12.0	23.8 ± 0.9		
					lu03-7	10	100–500	0.75	0.88	39.4	24.0 ± 0.4		
Lu05	6	4.4	55.2	2.4	lu05-4	10	300–460	0.88	0.88	37.6	30.4 ± 0.6	29.4 ± 2.1	5.4 ± 0.4
					lu05-5	5	300–460	0.36	0.74	4.4	26.8 ± 1.7		
					lu05-6	5	300–460	0.36	0.75	5.4	22.9 ± 1.1		
					lu05-7	9	100–460	0.74	0.83	22.6	27.8 ± 0.8		
Lu06	6	15.9	67.8	6.5	lu06-2	10	100–500	0.59	0.86	8.2	20.6 ± 1.3	22.8 ± 1.2	3.5 ± 0.2
					lu06-5	8	100–430	0.40	0.84	10.4	21.1 ± 0.7		
					lu06-7	6	300–500	0.48	0.79	28.9	23.3 ± 0.3		
Lu07	7	351.8	62.2	5.1	lu07-1	9	100–460	0.72	0.85	19.9	24.2 ± 0.7	24.0 ± 0.2	4.0 ± 0.0
					lu07-5	7	250–460	0.57	0.78	14.9	23.6 ± 0.7		
					lu07-8	8	100–430	0.47	0.84	8.6	23.8 ± 1.1		
Lu08	7	18.6	64.3	7.6	lu08-2	6	300–500	0.34	0.77	7.0	18.9 ± 0.7	25.5 ± 4.2	4.1 ± 0.7
					lu08-5	6	300–500	0.47	0.78	24.8	23.4 ± 0.3		
					lu08-6	10	100–500	0.72	0.88	21.6	33.1 ± 1.0		
Lu09	6	17.6	66.2	5.0	lu09-2	5	300–430	0.51	0.67	19.7	22.9 ± 0.4	23.9 ± 0.7	3.6 ± 0.1
					lu09-4	5	300–430	0.42	0.52	2.7	21.2 ± 1.7		
					lu09-5	9	100–460	0.37	0.70	5.5	23.0 ± 1.1		
Lu12	6	347.5	61.2	5.2	lu12-3	8	200–500	0.45	0.78	3.6	15.2 ± 1.5	14.4 ± 1.5	2.4 ± 0.3
					lu12-4	6	300–500	0.55	0.79	3.2	16.6 ± 2.3		
					lu12-5	6	300–500	0.58	0.75	2.4	11.5 ± 2.1		
Mean		359.8	62.2	3.8								23.0 ± 4.5	3.9 ± 0.9

demonstrate intraflow consistency. The weighted mean paleointensity values for these 7 flows range from 14.4 to 29.4 μT, and the corresponding VDM values range from 2.4 to 5.4 $\times$ 10^{22} Am^2. The mean VDM determined from all seven lava flows is (3.9 ± 0.9) $\times$ 10^{22} Am^2 (see Table 2).

5. Discussion and Conclusions

[10] Our study, employing the Thellier-Thellier absolute paleointensity method, provides 7 reliable determinations of geomagnetic field strength for the interval 23–22 Ma. The only other determinations presently available for this time interval were obtained by *Juarez et al.* [1998] on 22 ± 2 Ma submarine basalt glass (SBG). Yet the calculated mean VDM values for the two studies, (3.9 ± 0.9) $\times$ 10^{22} Am^2 and (4.4 ± 0.5) $\times$ 10^{22} Am^2, respectively, are remarkably similar.

[11] We now include the present result in the existing paleointensity database for the Cenozoic and, although there are relative few paleointensity estimates involved, attempt an analysis of paleofield strength for the time interval 65-7 Ma. In order for a given published set of determinations to be considered in our analysis the data are required to satisfy the following reliability criteria: (1) the determinations were obtained through some form of the step-wise Thellier-Thellier method with positive pTRM-checks; (2) at least two successful determinations per cooling unit were made available; (3) the standard error of the mean for a given

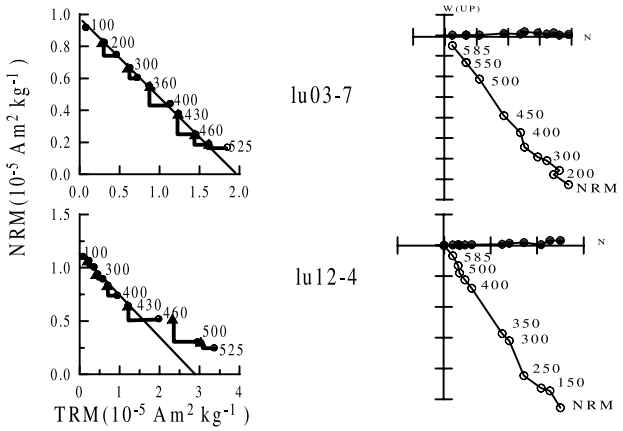


Figure 2. The representative NRM-TRM plots (left) and associated orthogonal diagrams (right) for Pingzhuang samples. In the NRM-TRM plots, filled (open) circles denote data points accepted (rejected) from the best-fit line used to define the paleointensity values, solid triangles are pTRM checks. In the orthogonal diagrams, The solid (open) symbols refer to the horizontal (vertical) plane. The numbers refer to temperatures in °C.

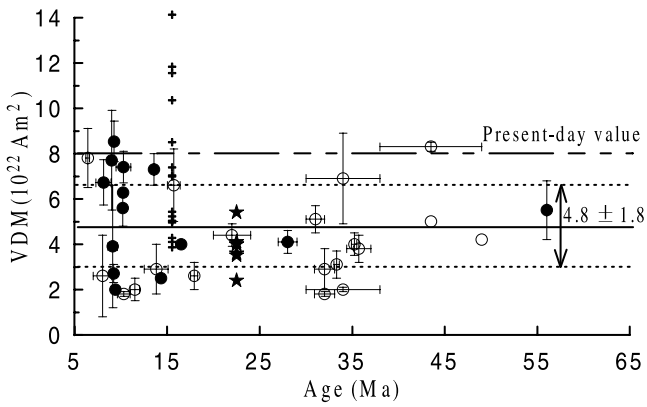


Figure 3. Collections of VDM data for 65-7 Ma. Solid circles represent data from IAGA database and several recent published papers (details see text), crosses represent data from Steens Mountain [*Prevot et al.*, 1985], open circles for data from SBG and stars are for this study. The slight lines are error bars for both VDM and ages.

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cooling unit must be less than 20%; and (4) the palaeointensities are not associated with a transitional field.

[12] The following datasets satisfy the above criteria: only one study [*Prévot et al.*, 1985] from the International Association of Geomagnetism and Aeronomy (IAGA) database (complied by *Mireille Perrin* and *Elisabeth Schnepp*, see also *Perrin et al.*, 1998); SBG study [*Juarez et al.*, 1998; *Selkin and Tauxe*, 2000]; and several recent studies [*Riisager et al.*, 1999, 2002; *Goguitchaichvili et al.*, 2000, 2001a, 2002; *Leonhardt et al.*, 2000; *Riisager and Perrin*, 2000]. When the Pingzhuang lava data are combined with these selected data, a mean VDM of $(5.26 \pm 2.67) \times 10^{22}$ Am^2 is obtained (Figure 3) for the entire 65-7 Ma interval, a value that is approximately half of that of the present day. However, SBG probably contains a grain-growth chemical remanent magnetization in magnetite, and are used to provide a lower bound estimate on intensities [*Heller et al.*, 2002].

[13] Similarly, *Prévot et al.* [1990] found that the average dipole field strength for the Mesozoic is about one-third of the present-day value. Termed the Mesozoic Dipole Low, the idea, however, is now seriously questioned [*Juarez et al.*, 1998; *Goguitchaichvili et al.*, 2002] though it has seen additional support [*Perrin and Scherbakov*, 1997]. Our compilation for the period 65-7 Ma (Figure 3), which now contains data from Inner Mongolia, finds the mean VDM to be $(5.26 \pm 2.67) \times 10^{22}$ Am^2, consistent with *Juarez et al.* [1998] assessment.

[14] Further, we note that the several paleointensity determinations in Figure 3 obtained from Steens Mountain [*Prévot et al*, 1985] show considerable variability with a VDM range from 4 to 14×10^{22} Am^2. However, many of these determinations correspond to the field strength either immediately before or after a reverse-to-normal Miocene transition that was recorded in detail. In particular, the data subsequent to the polarity change, which indicate both very strong and highly variable field strengths, may be more associated with the dynamo process of reversal and its return to "normalcy," than being spot readings relevant to non-transitional times. Indeed, strong paleointensities have been found following other Plio-Pleistocene polarity transitions [*Hoffman and Morse*, 1987; *Bogue and Paul*, 1994] though some argument raised [*Goguitchaichvili et al.*, 2001b]. Given this, if we now remove the data obtained from Steens Mountain and SBG from the 65-7 Ma dataset, which then gives a lowered mean field of $(4.8 \pm 1.8) \times 10^{22}$ Am^2. Hence, we propose that the Mesozoic-Cenozoic paleointensity data suggest that the higher strength of the modern-day field—rather than any low strength that may have existed during the Mesozoic—is the more geomagnetically anomalous over this span of geologic time. Clearly, more data are needed to better understand this long-term variation.

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References

Bogue, S. W., and H. A. Paul, Distinctive field behavior following geomagnetic reversals, *Geophys. Res. Lett.*, *20*, 2399–2402, 1994.

Coe, R. S., Palaeo-intensities of the earth's magnetic field determined from Tertiary and Quaternary rocks, *J. Geophys. Res.*, *72*, 3247–3262, 1967.

Coe, R. S., C. S. Grommé, and E. A. Mankinen, Geomagnetic palaeointensities from radiocarbon-dated lava flows on Hawaii and the question of the Pacific nondipole low, *J. Geophys. Res.*, *83*, 1740–1756, 1978.

Day, R., M. D. Fuller, and V. A. Schmidt, Hysteresis properties of titanomagnetites: grain size and composition dependence, *Phys. Earth Planet. Inter.*, *13*, 260–266, 1977.

Goguitchaichvili, A., L. M. Alva-Valdivia, J. Urrutia-Fucugauchi, and J. Morales, Absolute paleointensity results from the Trans-Mexican Volcanic Belt: Implications for the late Miocene geomagnetic field strength, *Geophys.J. Int.*, *143*, 977–984, 2000.

Goguitchaichvili, A., L. M. Alva-Valdivia, J. Urrutia-Fucugauchi, C. Zesati, and C. Caballero, Palaeomagnetic and palaeointensity study of Oligocene volcanic rocks from Chihuahua (northern Mexico), *Phys. Earth Planet. Inter.*, *124*, 223–236, 2001a.

Goguitchaichvili, A., C. Pierre, and J. Urrutia-Fucugauchi, On the features of the geodynamo following reversals or excursions: by absolute geomagnetic paleointensity data, *Phys. Earth Planet. Inter.*, *124*, 81–93, 2001b.

Goguitchaichvili, A., L. M. Alva-Valdivia, J. Urrutia, and J. Morales, On the stability of Mesozoic Dipole Low: New absolute paleointensity results from Parana Flood Basalts (Brazil), *Geophys. Res. Lett.*, *29*, 33-1–33-4, 2002.

Heller, R., R. T. Merrill, and P. L. McFadden, The variation of intensity of earth's magnetic field with time, *Phys. Earth Planet. Inter.*, *131*, 237–249, 2002.

Hoffman, K. A., and D. L. Morse, *EOS Trans. Am. Geophys. Union*, *68*, 1249, 1987.

Juarez, M. T., L. Tauxe, J. S. Gee, and T. Pick, The intensity of the Earth's magnetic field over the past 160 million years, *Nature*, *394*, 878–881, 1998.

Laj, C., and C. Kissel, Geomagnetic field intensity at Hawaii for the last 420 kyr from the Hawaii Scientific Drilling Project core, Big Island, Hawaii, *J. Geophy. Res.*, *104*(B7), 15,317–15,338, 1999.

Leonhardt, R., F. Hufenbecher, F. Heider, and H. C. Soffel, High absolute paleointensity during a mid Miocene excursion of the Earth's magnetic field, *Earth Planet. Sci. Lett.*, *184*, 141–154, 2000.

Perrin, M., and V. Shcherbakov, Palaeointensity of the Earth's magnetic field for the past 400Ma: Evidence for a dipole structure during the Mesozoic low, *J. Geomagn. Geoelectr.*, *49*, 601–614, 1997.

Perrin, M., E. Schnepp, and V. Shcherbakov, Palaeointensity database updated,, *EOS*, *79*, 16, 1998.

Prévot, M., M. E. M. Derder, M. McWilliams, and J. Thompson, Intensity of the Earth's magnetic field: evidence for a Mesozoic dipole low, *Earth Planet. Sci. Lett.*, *97*, 129–139, 1990.

Prévot, M., E. A. Mankinen, R. S. Coe, and C. S. Gromme, The Steens Mountain (Oregon) geomagnetic polarity transition, 2. Field intensity variations and discussion of reversal models, *J. Geophys. Res.*, *90*, 10,417–10,448, 1985.

Riisager, J., M. Perrin, and P. Rochette, Palaeointensity results from Ethiopian basalts: implications for the Oligocene geomagnetic field strength, *Geophys. J. Int.*, *138*, 590–596, 1999.

Riisager, J., and M. Perrin, Palaeomagnetism, palaeointensity and geochronology of Miocene basalts and baked sediments from Velay Oriental, French Massif Central, *J. Geophys. Res.*, *105*, 883–896, 2000.

Riisager, P., J. Riisager, N. Abrahamsen, and R. Waagstein, Thellier palaeointensity experiments on Faroes Flood Basalts: Technical aspects and Geomagnetic Implications, *Phys. Earth Planet. Inter.*, *4102*, 1–10, 2002.

Selkin, P. A., and L. Tauxe, Long-term variations in palaeointensity, *Phil. Trans. R. Soc. London A*, *358*, 1065–1088, 2000.

Thellier, E., and O. Thellier, Sur l'intensite du champ magnetique terrestre dans le passé, historique et geologique, *Ann. Geophys.*, *15*, 285–376, 1959.

Zhao, X., R. Coe, Y. Zhou, S. Su, H. Wu, G. Kuang, Z. Dong, and J. Wang, Tertiary palaeomagnetism of North and South China and a reappraisal of late Mesozoic palaeomagnetic data from Eurasia: implications for the Cenozoic tectonic history of Asia, *Tectonophysics*, *235*, 181–203, 1994.

Zhu, R., Y. Pan, J. Shaw, D. Li, and Q. Li, Geomagnetic palaeointensity just prior to the Cretaceous Normal Superchron, *Phys. Earth. Planet. Inter.*, *128*(1–4), 207–222, 2001.

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